

Pranay Grandhi

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PROFESSIONAL SUMMARY

Computer Science and Cyber Security undergraduate with a strong foundation in cybersecurity, network security, vulnerability assessment, web application security, authentication and authorization, encryption fundamentals, security monitoring, incident response concepts, and secure system design. Hands-on experience in vulnerability assessment, penetration testing, network traffic analysis, attack identification, and security analysis using Wireshark, Nmap, Burp Suite, and Linux. Knowledge of TCP/IP, DNS, HTTP/HTTPS, firewalls, common network threats, OWASP Top 10, security controls, and security best practices. Strong analytical, problem-solving, research, documentation, and communication skills with a strong interest in application, system, and network security.

EDUCATION

SRM Institute of Science and Technology <i>B.Tech – Computer Science and Engineering (Cyber Security)</i>	2024 – Present <i>CGPA: 9.33</i>
Intermediate (Class XII)	83%
Secondary School (Class X)	93.1%

EXPERIENCE

Network Security Associate Intern – Fortinet <i>EduSkills</i>	Jul 2024 – Oct 2024
- Analyzed network traffic and identified potential security vulnerabilities using industry-standard security tools. - Applied firewall, IDS, network security, TCP/IP, and basic network security concepts to strengthen security analysis capabilities. - Studied network configurations, security controls, common network threats, and attack patterns in security-focused environments.	
Ethical Hacking Intern <i>EduSkills</i>	Jul 2025 – Oct 2025
- Conducted vulnerability assessments and penetration testing in simulated and authorized environments to identify security weaknesses. - Applied ethical hacking methodologies to analyze vulnerabilities and evaluate application and system security. - Identified security weaknesses and proposed mitigation and remediation strategies following security best practices. - Applied security concepts including authentication, authorization, input validation, secure coding, and web application security during security testing and analysis.	

CYBERSECURITY PROJECTS

Security Analysis of Internet-Based Voice and Video Communication Infrastructure
- Analyzed SIP and RTP VoIP protocols across 100+ packets using Wireshark for network and security analysis. - Performed packet-level traffic analysis to identify unusual communication patterns and security weaknesses. - Identified Man-in-the-Middle (MITM) and replay attack scenarios through protocol and network traffic analysis. - Implemented Secure RTP (SRTP) and TLS encryption to improve communication security. - Evaluated secure communication mechanisms and encryption fundamentals for protecting network traffic.
Practical Attacks and Countermeasures for Cryptosystems
- Simulated brute-force, timing, and cryptographic attacks to analyze vulnerabilities in cryptographic systems. - Designed security countermeasures using robust key management and encryption techniques. - Evaluated attack scenarios and recommended security improvements to reduce vulnerability exposure.
Security Analysis of CPU–GPU Coupled Architectures
- Investigated security vulnerabilities in heterogeneous CPU–GPU architectures. - Identified memory leakage and side-channel attack risks affecting system security. - Proposed secure memory isolation techniques to improve system resilience and reduce security exposure.

ADDITIONAL PROJECT EXPERIENCE

IN-SPACE Rocketry Student Competition – Critical Design Review Qualifier
- Qualified for the Critical Design Review (CDR) after clearing the Preliminary Design Review (PDR) in the national-level IN-SPACE Rocketry Student Competition. - Collaborated with a multidisciplinary engineering team on the design and development of a student-built sounding rocket. - Contributed to subsystem design, system integration, engineering calculations, and technical documentation. - Participated in design validation, risk assessment, trade-off analysis, and preparation of technical reports and presentations.

TECHNICAL SKILLS

Cybersecurity: Vulnerability Assessment, Penetration Testing, Ethical Hacking, Web Application Security, Network Security, System Security, Threat Analysis, Security Controls, Security Monitoring, Incident Response Concepts, Information Security
Network Security: TCP/IP, DNS, HTTP/HTTPS, Firewalls, IDS, Network Traffic Analysis, Common Network Threats,

Attack Patterns, Secure Communication

Application Security: Authentication, Authorization, Input Validation, Secure Coding Fundamentals, OWASP Top 10, Web Application Security

Security Tools and Platforms: Linux, Wireshark, Nmap, Burp Suite, Git, GitHub, Visual Studio Code

Security Concepts: Encryption Fundamentals, Cryptography, Vulnerability Management, Security Best Practices, Risk Assessment, Security Analysis

Programming Languages: Python, C, C++, Java

SOFT SKILLS

Analytical Thinking | Problem Solving | Attention to Detail | Security Research | Technical Documentation
| Team Collaboration | Communication | Continuous Learning | Critical Thinking | Time Management

CERTIFICATIONS

Fortinet Certified Associate (FCA) | Fortinet Certified Fundamentals (FCF) | Cisco Network Defence | ISO/IEC 27001:2022 Lead Auditor